

ARITRI HALDER

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Mechanical Engineer | Thermal systems | GD&T | Prototype Builds | Test & Validation | NPD/NPI | Data Analysis

SUMMARY

Ph.D. candidate in Mechanical Engineering with hands-on R&D experience with DFMA, V&V, NPD. Designed and tested a novel 20 kW high-temperature industrial burner, integrated sensors and controls, validated system performance, and used experimental data to guide engineering decisions and achieved 60% reduction in NOx emissions compared to conventional burners.

EDUCATION

University of Minnesota - Twin Cities

Minneapolis, MN | Expected June 2027

Ph.D., Mechanical Engineering

M.S., Mechanical Engineering

Minneapolis, MN | October 2026

Jadavpur University

Kolkata, India | July 2023

B.E., Mechanical Engineering

EXPERIENCE

University of Minnesota - T.E. Murphy Engine Lab | Graduate Research Assistant

Minneapolis, MN | Oct 2023 - Present

- Designed and validated a **novel 20 kW High-temperature Furnace** and modular test hardware, delivering repeatable 100% ammonia combustion and achieving **60% reduction** in NOx.
- Developed SolidWorks CAD models**, GD&T drawings, and BOMs of swirl vanes and nozzle geometry to **maintain symmetric flow** and improve combustion performance.
- Selected** stainless-steel wetted hardware and PTFE seals for **ammonia compatibility** and operation near 1000 deg C, supporting safe and chemically compatible gas delivery.
- Planned and executed** a four-variable DOE and applied **advanced optical and laser diagnostics**, (FTIR, PIV, chemiluminescence, and CT mechanism) to define stable operating windows, validate performance, and characterize NOx/ammonia trends.
- Built and aligned** a low-speed 2D-2C PIV **optical system** with convex, concave, and cylindrical lenses on translation stages; optimized beam collimation and lens spacing to produce a 0.5 mm thick laser sheet and **quantify velocity measurements**.
- Aligned** a single-camera rotational **CT mechanism** for multi-angle flame imaging and developed custom image-processing algorithms in Python/MATLAB to quantify burner process behavior and **iterate design improvements**.
- Engineered and commissioned an ammonia/methane/air process system integrating mass flow controllers, sensors, NI cDAQ, and LabVIEW for **automated control and synchronized data acquisition**.
- Troubleshoot** gas delivery, controls, instrumentation, and diagnostics **using RCA/FMEA**; implemented corrective changes to improve test reliability and safety.
- Collaborated with CenterPoint Energy and U of M Duluth to validate system performance and **translate experimental results into scaled up industrial application**.

IIT Kanpur - Experimental Thermofluids Laboratory | Research Intern

Kanpur, India | May 2022 - Jul 2022

- Ran **parameterized study** of mesh convergence **to verify the pressure** results were independent of the mesh size in **ANSYS Fluent**.
- By modifying internal insert geometry and CFD simulation results **improved** Minichannel Heat Sink Thermal-fluid performance by **up to 28%**.

SELECTED PUBLICATION & PRESENTATIONS

- A. Halder**, C. Goertemiller, A. Hoxie, and W. F. Northrop, "Experimental investigation of a 20 kW atmospheric pressure pure ammonia jet-swirl burner," *Proceedings of the Combustion Institute*, 2026. [[Link](#)]
- A. Halder**, R. Pandey, and U. Madanan, "Computer simulations on heat transfer enhancement in a minichannel heat sink using modified butterfly inserts," *ISHMT-ASTFE Heat and Mass Transfer Conference*, pp. 149–154, 2024. [[Link](#)]
- A. Halder et al.**, "Stability and Emissions Profile of a 120 kW Fuel-Flexible Jet-Swirl Burner Operating on Ammonia." In progress; *Transformative Energy*.
- Five oral/poster presentations at international and **U.S. technical meetings**, including the **41st International Symposium on Combustion** (ISOC Kyoto, Japan, 2026), 14th U.S. National Combustion Meeting (Boston, MA, 2025), and 4th Symposium on Ammonia Energy (2025).

SKILLS

Mechanical Design & Precision Hardware: SolidWorks, engineering drawings, GD&T, tolerance stack-up, concentricity/alignment, fit/clearance analysis, DFM/DFA, BOMs, materials selection, prototyping, AutoCAD, Fusion 360.

Opto-Mechanical & Diagnostics: 2D-2C PIV, laser-sheet optics, convex/concave/cylindrical lenses, translation/rotation stages, optical alignment, camera calibration, reflection troubleshooting, FTIR, chemiluminescence, CTC.

Instrumentation & Integration: NI cDAQ, LabVIEW, MFCs, pressure/temperature sensors, synchronized DAQ, test automation, ammonia/methane/air systems, calibration, measurement validation.

Analysis & Problem Solving: ANSYS Fluent, CFD, FEM, heat transfer, thermal-fluid modeling, DOE, FMEA, RCA, statistics, Python, MATLAB, C/C++, Tecplot, HPC, Cantera.

AWARDS

- Chester E. Dekko Fdn Endowed Graduate Fellowship**, University of Minnesota, 2024 - 2025.
- Best Abstract - Thermal Applications**, 4th Symposium on Ammonia Energy, 2025.